

18-April-2026

Agenda: Statistics -9

: z-score, p-value

: CLT, SE, significance level, Z-stats, T-test, Hypothesis testing

: confidence interval

Z-score: (universal translator)

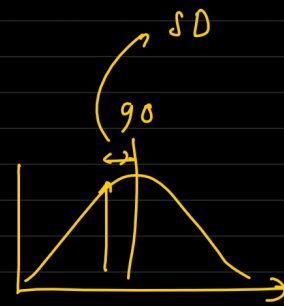
→ Zscore tells how far a point is from the mean,

→ unit of Zscore: Standard Deviation

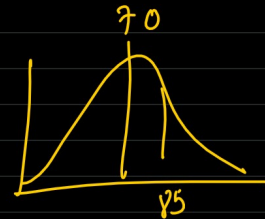
example:

scored = 85 marks $\begin{cases} \text{history} \\ \text{math} \end{cases}$

math $\rightarrow$ student $\rightarrow$ average 90 $\rightarrow$
 $\rightarrow$ low



history $\rightarrow$ student $\rightarrow$ 70 $\rightarrow$
 $\rightarrow$ 85



$$Z_{\text{score}} = \frac{x - \mu(\text{mean})}{\sigma(\text{SD})} = \frac{x - \mu}{\sigma}$$

P.1

$\sigma = 10\text{cm}$

$\mu = 150\text{cm}$

$x = 170\text{cm}$

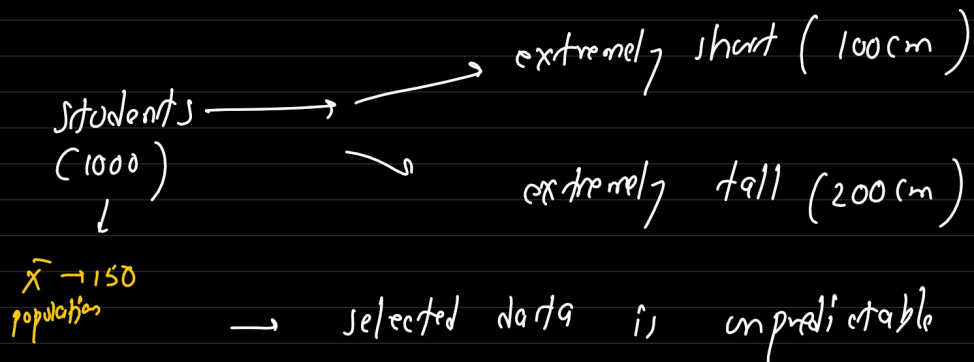
$$= \frac{170 - 150}{10} = \frac{20}{10} = 2$$

area under curve prob $\rightarrow$ Z-table

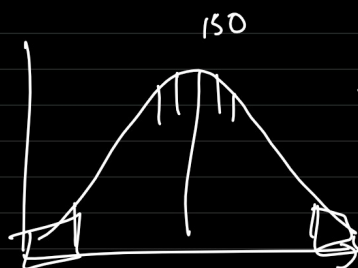
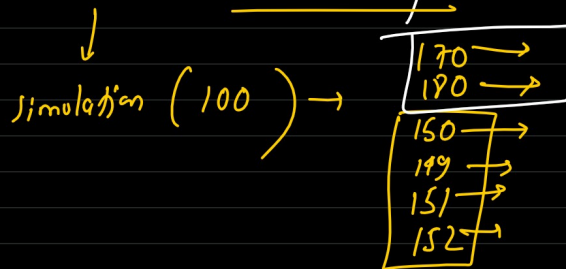
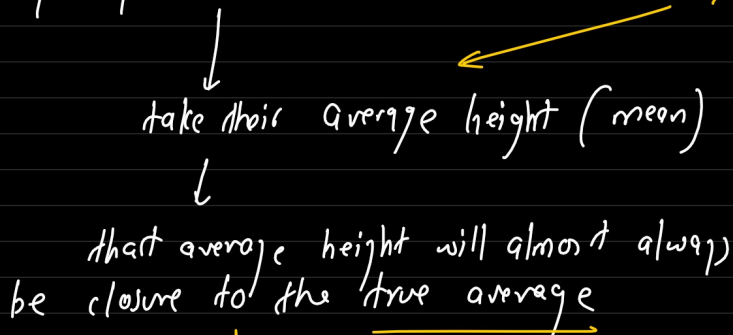
2 $\rightarrow$ 0.9772
 $\rightarrow$ 97.72%

97% taller than your peers.

central limit theorem:



However, if you pick 100 random students $\rightarrow$ group



$\rightarrow$ average of group, will also form normal distribution

sample size $n \geq 30$

$\downarrow$ no of data points.

Standard Error (SE)

SD: Measure the spread of individuals from the mean.

SE: Measure the spread of group average from the mean.

SD: How much do single kids vary?

SE: How much do averages of 40 kids vary?



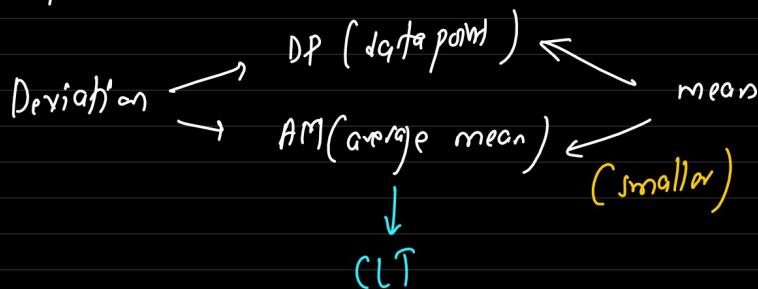
① mean $\leftarrow$ SD $\rightarrow$

② mean $\leftarrow$ SE $\rightarrow$



DP $\rightarrow$ Data point

average mean $\rightarrow$ mean $\rightarrow$ predicted $\rightarrow$



SD $\rightarrow$ 2

SE < SD $\rightarrow$

$$SE = \frac{\sigma}{\sqrt{n}}$$

Scenario: group of 25 students, $SD(\sigma) = 10$ cm

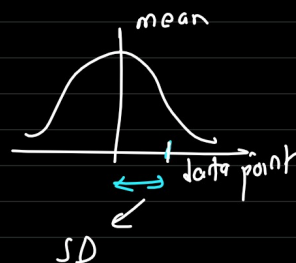
$$SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2$$

z-score = $\frac{\text{value} - \text{mean}}{SD}$ $\rightarrow$ measuring an individual

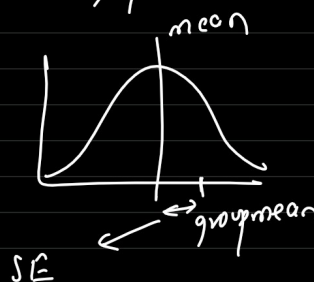
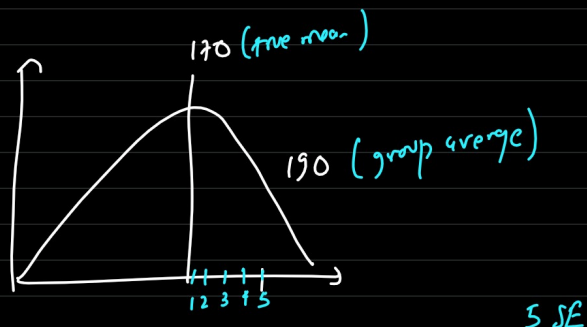
z-statistic = $\frac{\text{sample mean} - \text{population}}{\text{standard error}}$

$$= \frac{\bar{x} - \mu}{SE} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

z-score: how many SD fit into the gap.



z-statistic: how many standard error fit into the gap.



$$Z_{stat} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Scenario:

The claim: A company says that their energy drink has 200mg caffeine on average.

The evidence:

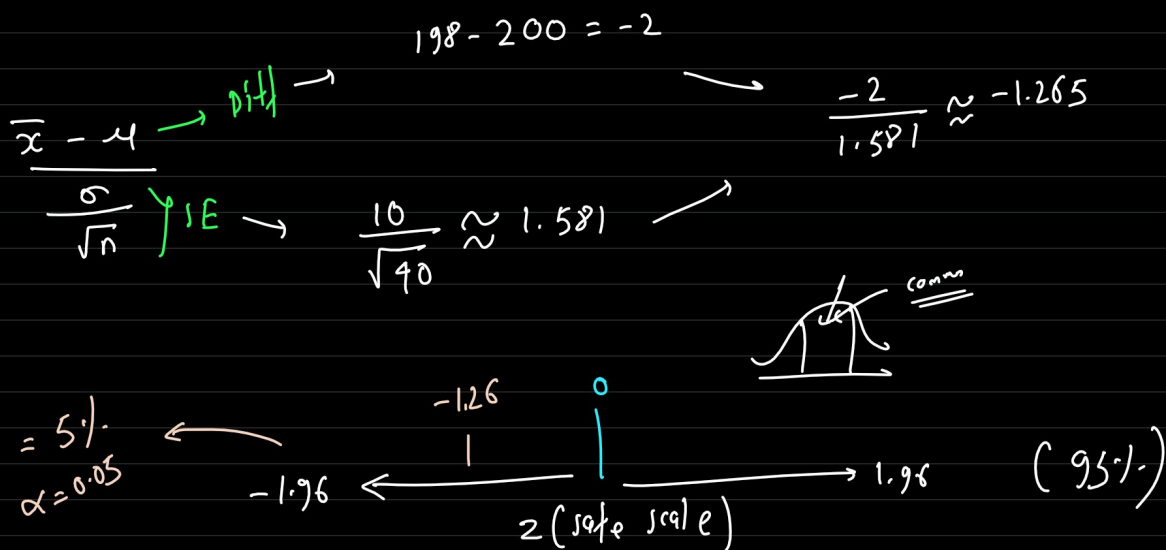
population mean (μ) = 200mg

population deviation (SD or σ) = 10mg

sample size (n) = 40 cans

sample mean ($\bar{x}$) = 198mg $\rightarrow$

Question: Is our 198mg result "significantly different" or just lucky/unlucky/random sample?

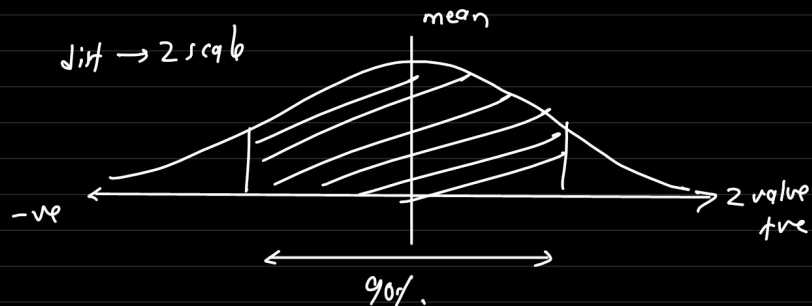


198 is lower than 200, but it's not weird enough to prove that company is lying. It's probably just a random coincidence.

90%

95%

99%



	0.01	0.02	0.03	0.04	...	0.09
1.0						
1.1						
1.2						
1.3						
1.4						

1.39

1.39

prob.

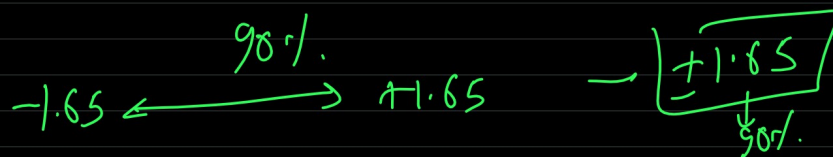
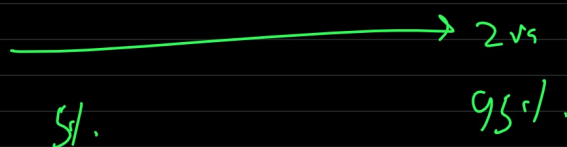
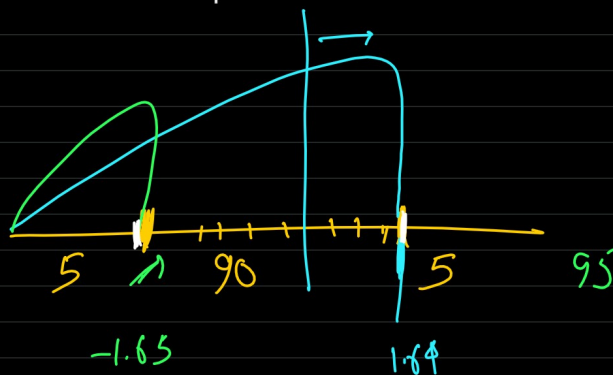
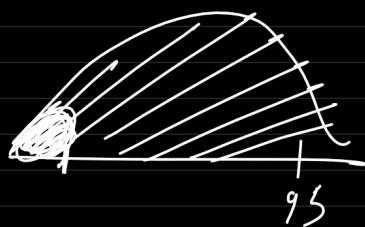
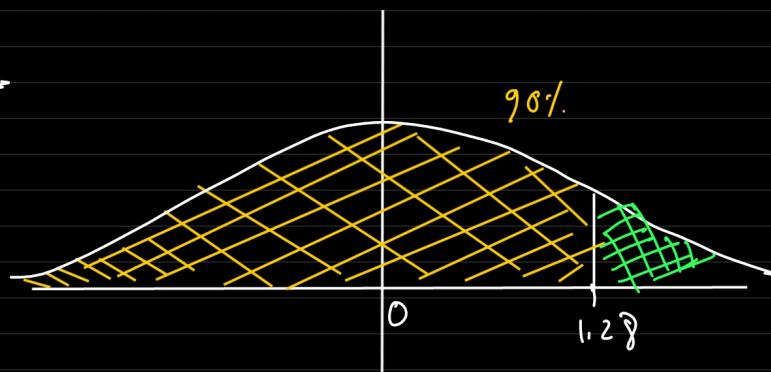
1.39

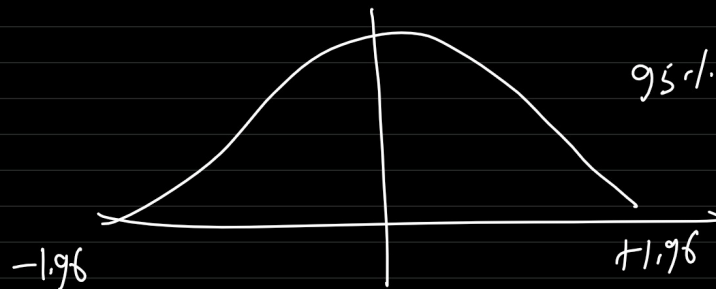
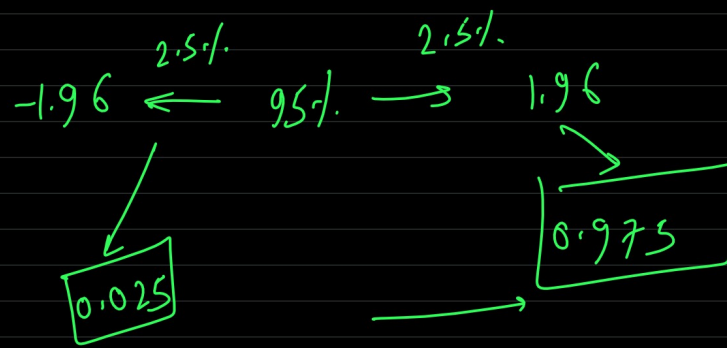
1.39

0.09

90%

900





Significance - level (α)

A $\rightarrow$ flip coin 2 times $\rightarrow$ 2 heads

$\rightarrow$ flip 20 times $\rightarrow$ 20 heads

$\nearrow$
This is too unlikely to be luck

$\downarrow$
too rare

$\rightarrow$ math rule $\rightarrow$ significance level,

$\rightarrow$ The significance level (α) is the probability of rejecting the claim when it is actually true.

$\rightarrow$ Tolerance for error

$\alpha = 0.05 \rightarrow 5\% \rightarrow$ we are okay with being wrong
5% of the time

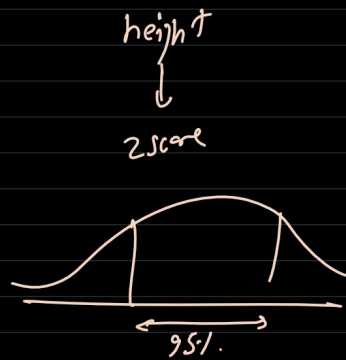
$\alpha = 0.01$ (1%) : we only want to be wrong 1% of the time.

$$\alpha = 0.05 \text{ (5\%)}$$

95% sure

$$\alpha = 0.05 \rightarrow 5\%.$$

95%.



Significance level

Two tailed

confidence level

$$\alpha = 0.05 \rightarrow z \rightarrow \pm 1.96 \rightarrow 95\%$$

$$\alpha = 0.01 \rightarrow z \rightarrow \pm 2.58 \rightarrow 99\%$$

$$\alpha = 0.10 \rightarrow z \rightarrow \pm 1.645 \rightarrow 90\%$$

Hypothesis testing!

↓
court

↓
innocent until proven guilty

Null hypothesis (H_0) = "innocent state"
= "everything is normal"

Alternate hypothesis (H_1) = "guilty state"
= "accusation"
= "something has changed!"
"You are slower!"

Tail:

Two-Tailed Test:

: You don't care if it's faster or slower, you just want to know if it's not 30 minutes

: 5% significance $\rightarrow$ left $\rightarrow$ 2.5%
right $\rightarrow$ 2.5%

: critical value: ± 1.96

One-tailed test:

: The customer group specifically claims you are slower.

$\downarrow$
30 $> 30 \rightarrow$ slower

 $\downarrow$
we only care about right side of the curve.

: 5% significance

: critical value: $+1.645$

P.):

The claim: average delivery time ($\mu = \underline{30}$ minutes)
(H_0)

The Accusation: A customer group says you are slower,
(H_1):

They tested 36 deliveries ($n = 36$)

Evidence: $\bar{x} = 31$ minutes

population standard deviation (σ) = 4 minutes

significance level (α) = 0.05 (95% confidence)

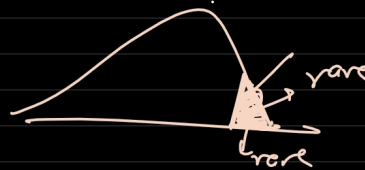
step 1: state the hypothesis

$$H_0: \mu = 30$$

$$H_1: \mu > 31 \quad (\text{one tailed alternative})$$

step 2: set the decision rule, $\alpha = 0.05$

group average $\rightarrow$ ~~z-score~~
 $\rightarrow$ z-statistic



$$z\text{-statistic} > 1.645,$$

"outrage"
 $\downarrow$
reject H_0

$H_1 \rightarrow$ fail to reject H_0
 $\rightarrow$ reject H_0

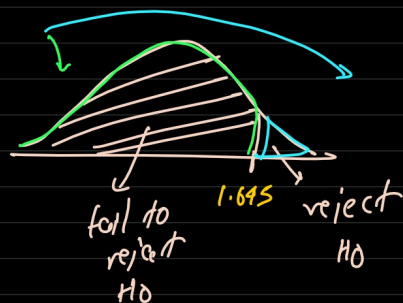
step 3: calculate SE

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{36}} = 0.6667$$

step 4: calculate z-statistic

$$z_{\text{stat}} = \frac{\bar{x} - \mu}{SE} = \frac{31 - 30}{0.6667} = \frac{1}{0.6667} \approx 1.50$$

$$z_{\text{stat}} = 1.50$$



$$\text{is } 1.50 > 1.645 \rightarrow \text{No}$$

conclusion: Fail to reject Null Hypothesis

Even though the average was 31 minutes, a sample of 36 is small enough that this could have just been a bad week by random chance.

we don't have enough proof to call the company.

p-value

A → flipped coin 2 times → 2 heads
flipped coin 10 times → 10 heads

0.1% chance → how sure we are

↑
p-value

p-value → lower p-value means rare
→ higher p-value means common

→ $[0, 1]$

Rule: if the p is low, the Null Hypothesis is rejected

$(p\text{-value} < \alpha, \text{ we reject the } H_0)$

Example:

$$H_0 = \mu = 30$$

$$H_1 = \mu > 30 \text{ (one tailed)}$$

$$z\text{-stats} : 1.5$$

$$\text{Alpha}(\alpha) : 0.05$$

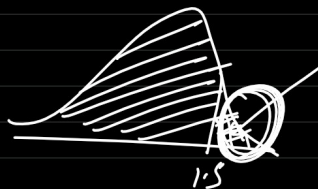
$$P(Z > 1.50)$$

$$\rightarrow z = 1.50 \rightarrow \text{Area?}$$

$$\rightarrow 0.93319$$

$$\rightarrow 0.9332$$

$$\rightarrow 93.32\% \text{ change the result is less than } 1.50$$



$$p\text{-value} = 1 - \text{Area to the left}$$

$$= 1 - 0.9332$$

$$= 0.0668$$

$$\alpha = 0.05$$

$$0.0668 < 0.05$$

$$6.68\% < 5\%$$

there's a 6.68 chance that our 31-minute delivery average was a random fluke.

= Not statistically significant.